Name: ____________________________ Partner: _______________________ Period: _________

Tier R — Remediate

Directions: For each equation below, identify the type of conic, the center or vertex, and the key measurements. No equation-writing is required.

Card A: $y - 4 = 3(x+2)^2$
Type: _______________ Vertex: __________________ $a =$ _________ Opens: ____________

Card B: $\dfrac{(x-3)^2}{25} + \dfrac{(y+1)^2}{9} = 1$
Type: _______________ Center: __________________ $a =$ _________ $b =$ _________

Card C: $(x+4)^2 + (y-2)^2 = 36$
Type: _______________ Center: __________________ $r =$ _________

Card D: $\dfrac{(x-1)^2}{9} - \dfrac{(y-2)^2}{4} = 1$
Type: _______________ Center: __________________ $a =$ _________ $b =$ _________ Opens: ____________

Tier A — Approaching Proficiency

Directions: Write the standard form equation for each card below. Show all work.

Card A: "An equation that opens to the right with vertex at $(-1, 2)$ and passes through $(3, 4)$." (Hint: substitute the point to find a.)

Card B: "An ellipse centered at the origin with horizontal radius 4 and vertical radius 3."

Card C: "A hyperbola centered at $(1, 0)$ with $a = 3$ and $b = 2$, opening left and right."

Card D: "A circle centered at $(-2, 5)$ with radius 3."

Tier E — Extension

Part 1. For Card B (Tier A), find all x-intercepts and y-intercepts of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Show all work.

Part 2. For Card C (Tier A), write both asymptote equations for the hyperbola $\frac{(x-1)^2}{9} - \frac{y^2}{4} = 1$.

Part 3. Create your own Card E. Choose a conic type, state its key features in words (as if writing a card description), then write its equation.

My Card E (description): ____________________

My Card E (equation): ____________________